

Daytime Running Lights Controller

T.K. HAREENDRAN

Auto makers have been gradually switching to light emitting diode (LED) lighting for automotive headlamps because of its features such as high efficiency and long service life. In addition, from a safety perspective, applications of LED-driven daylight/daytime running lights (DRLs) for vehicles are spreading in many states.

The purpose of the circuit presented here is to activate DRLs on any lighting that uses LED and/or in-

candescent bulb in a vehicle. Before attempting to construct this circuit, remember that, you cannot directly hook up the circuit to any circuit that is controlled by the CANbus system in a vehicle. For example, if the parking lights of your vehicle are CANbus-controlled, the DRL circuit cannot be plugged to the parking-light circuit for DRL function.

But, if the fog-light circuit is not controlled by CANbus, then you can connect the DRL circuit to it. Author's prototype is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the DRL controller. It is built around

timer NE555 (IC1), MOSFET 60NF06 (IRF1), 12V, 1C/O relay (RL1), DRLs and a few other components.

There are seven wires that come out of the circuit. The first connection (DRL-B and DRL-G) you will make is to the DRLs. These are the main wires that will make the bumper DRLs turn on when you start the vehicle (these will light up at start).

Connect DRL-B and DRL-G wires from the circuit directly to the DRLs at the bumper. The circuit activates when it senses ignition voltage. It does so by getting a signal from the main wire (IGN +) and the positive

supply wire that runs from the circuit to the ignition-switched +12V power line. GND is main ground connection, and it must be connected directly to the negative battery (0V) terminal or the body of the vehicle.

You might have to extend the wire, if

it does not reach the battery, by running sufficient length of the automotive wire from the circuit to the negative terminal of the battery. If you want the DRLs to switch off when you turn your headlights and/or parking lights on, connect HL+ and PL+ to the existing headlight and parking-light wires, respectively.

Wire connection PB+ is optional; you do not have to connect it unless you want the DRLs to work with the parking brake (hand brake). The potmeter (VR1) can be used to adjust the brightness of DRLs as per requirement. Note that, you can modify the circuit's default

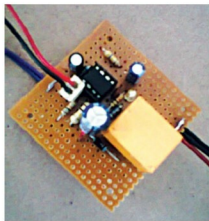


Fig. 1: Author's prototype

Wiring Guide Table

Signal	Remarks
DRL-G	Negative (0V) of daytime running light
DRL-B	Positive (12V) of daytime running light
IGN+	Positive (12V) supply from ignition switch
HL+	Positive (12V) supply from headlight wiring
PL+	Positive (12V) supply from parking-light wiring
PB+	Positive (12V) supply from parking-brake/hand-brake wiring
GND	Common ground (0V) connection/vehicle body

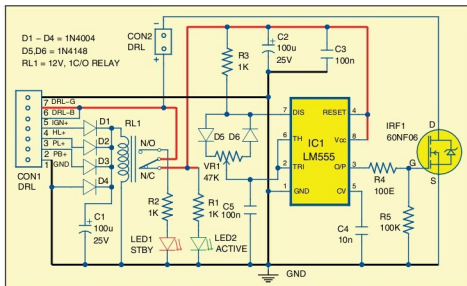


Fig. 2: Circuit diagram of the DRL controller

